Chapter 3

Viscous Effects in Aerodynamic Flows

This chapter will examine the changes in an aerodynamic flow caused by the presence of viscous wall boundary layers and trailing wakes. The objective is to model and quantify these changes and to explain their associated phenomena such as loss of lift at stall.

3.1 Inviscid Flow Model

Lumping of the vorticity in the viscous layers into vortex sheets, as illustrated in Figure 2.12 in Chapter 2, produces a fictitious strictly-potential *Equivalent Inviscid Flow* (EIF) velocity field $u_{\mathrm{i}}, v_{\mathrm{i}(s,n)}$. The EIF also has a pressure field $p_{\mathrm{i}(s,n)}$ related to $u_{\mathrm{i}}, v_{\mathrm{i}}$ by the Bernoulli equation (1.109).

$$p_{\mathrm{i}(s,n)} \;=\; p_\infty + \frac{1}{2}\rho V_\infty^2 \;-\; \frac{1}{2}\rho(u_{\mathrm{i}}^2 + v_{\mathrm{i}}^2) \tag{3.1}$$

Here u, v will denote velocity components along the local orthogonal sheet coordinates s, n.

Figure 3.1 compares the EIF to the real flow in more detail, for the case where the lumped vortex sheet is placed on the real surface. It is labeled "Simple" to distinguish it from the more advanced and accurate models considered later, which mostly eliminate the modeling discrepancies in the velocity and pressure in the Simple model.

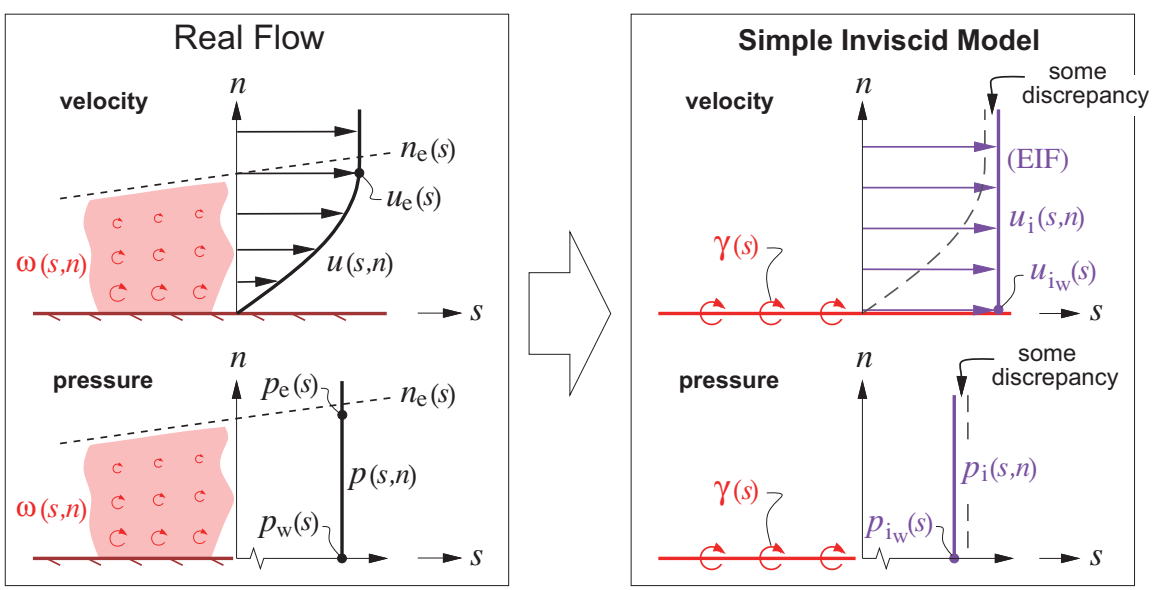

Figure 3.1: Real viscous flow approximated by Simple Inviscid Model. The v and v_{i} wall-normal velocity components are small and are not shown. This is a zoom-in of Figure 2.12.

Since the real flow in the boundary layer has almost parallel streamlines, the static pressure across it must be almost constant regardless of the velocity distribution. Hence the wall pressure $p_{\mathrm{w}}(s)$ is nearly equal to the edge pressure $p_{\mathrm{e}}(s)$, which in turn is related to the edge velocity via the Bernoulli equation. The negligibly small wall-normal velocity contribution $v_{\mathrm{e}}^2 \ll u_{\mathrm{e}}^2$ is omitted.

$$p_{\mathrm{w}}(s) \;\simeq\; p_{\mathrm{e}}(s) \;=\; p_\infty + \frac{1}{2}\rho V_\infty^2 \;-\; \frac{1}{2}\rho u_{\mathrm{e}}^2 \tag{3.2}$$

The subscript $(\;)_{\mathrm{e}}$ will in general denote an "edge" quantity in the irrotational flow just outside the edge of viscous layer, which is demarked by the $n_{\mathrm{e}}(s)$ curve shown in Figure 3.1. The subscript $(\;)_{\mathrm{w}}$ will denote a wall quantity at $n=0$.

If the boundary layer is thin the vorticity-lumping procedure will incur little error, in which case the wall velocity of the EIF closely approximates the edge velocity of the real flow.

$$u_{\mathrm{i_w}}(s) \;\simeq\; u_{\mathrm{e}}(s) \tag{3.3}$$

Combining (3.1),(3.2),(3.3) we have

$$p_{\mathrm{i_w}}(s) \;\simeq\; p_{\mathrm{w}}(s) \tag{3.4}$$

so that the EIF captures the real surface pressures, and hence should produce reasonably accurate lift forces and moments. Of course the EIF cannot represent the viscous skin friction in the real flow, so that it cannot correctly predict the drag. This will be addressed in Chapter 4.

Although the above discussion assumed incompressible flows, the EIF concept and flow models can be applied to compressible flows. The only differences are that the compressible Bernoulli relation (1.112) would be used in lieu of (3.1), and a grid method would be used in lieu of the vortex-sheet EIF model shown in Figure 3.1.

3.2 Displacement Effect

3.2.1 Normal mass flux matching

The major shortcoming of the simple inviscid model shown in Figure 3.1 is that it does not account for the *displacement effect* of the slower-moving fluid inside the boundary layer. This acts as a wedge, tilting and displacing the outer streamlines away from the wall, as shown in Figure 3.2.

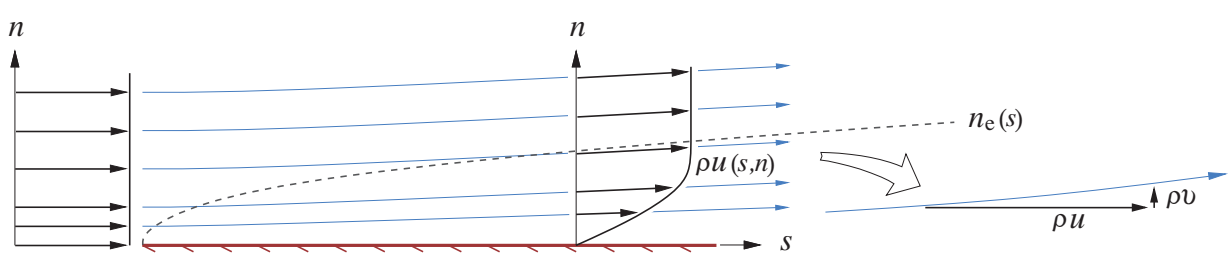

Figure 3.2: Actual viscous flow with displaced streamlines and corresponding vertical mass flux ρv, caused by piling up of the slower-moving fluid in the boundary layer.

This displacement changes the apparent flow tangency seen by the bulk flow, and thus modifies the overall flow-field. In the simple inviscid model this effect is ignored, which is the main reason for the discrepancies between the EIF's and real flow's edge velocity, wall pressure, and lift, shown in Figure 3.1. If the boundary layers are thin, then these discrepancies are small and are often ignored. But if the boundary layers are thick,

perhaps due to a low Reynolds number or the airfoil being close to or beyond stall, then the discrepancies between the real flow and the Simple Inviscid Model's EIF may be unacceptably large.

To mostly eliminate these modeling errors, the EIF must be constructed so that its vertical mass flux equals that of the real flow outside the real boundary layer.

$$\boxed{\rho v_{(s,n)} \;=\; \rho_{\rm i} v_{{\rm i}(s,n)} \qquad (\text{ for } n > n_{\rm e}(s)\ )} \tag{3.5}$$

3.2.2 Normal mass flux in real flow

Consider the real viscous flow shown in Figure 3.2. The wall-normal mass flux ρv at some location $n > n_{\rm e}$ outside the shear layer is computed by integrating its n-gradient from the wall. After also invoking mass continuity, the result is more concisely given in terms of the shear layer's mass defect $m(s)$, or the related displacement thickness $\delta^*(s)$.

$$\begin{aligned}
\rho v_{(s,n)} &= \rho v_{(s,0)} + \int_0^n \frac{\partial\, \rho v}{\partial n}\, {\rm d}n \;=\; 0 + \int_0^n -\frac{\partial\, \rho u}{\partial s}\, {\rm d}n \\
&= \int_0^n \frac{\partial}{\partial s}\left(\rho_{\rm e} u_{\rm e} - \rho u\right)\, {\rm d}n \;-\; n\,\frac{{\rm d}\,\rho_{\rm e} u_{\rm e}}{{\rm d}s} \\
&= \frac{\rm d}{{\rm d}s}\left[\rho_{\rm e} u_{\rm e} \int_0^{n_{\rm e}} \left(1 - \frac{\rho u}{\rho_{\rm e} u_{\rm e}}\right) {\rm d}n\right] \;-\; n\,\frac{{\rm d}\,\rho_{\rm e} u_{\rm e}}{{\rm d}s} \\
\text{or} \qquad \rho v_{(s,n)} &= \frac{{\rm d}m}{{\rm d}s} \;-\; n\frac{{\rm d}\,\rho_{\rm e} u_{\rm e}}{{\rm d}s} \qquad\qquad (\text{actual flow, for } n > n_{\rm e}) \\
\text{where} \qquad m(s) &\equiv \int_0^{n_{\rm e}} (\rho_{\rm e} u_{\rm e} - \rho u)\, {\rm d}n \;=\; \rho_{\rm e} u_{\rm e} \delta^* \\
\delta^*(s) &\equiv \int_0^{n_{\rm e}} \left(1 - \frac{\rho u}{\rho_{\rm e} u_{\rm e}}\right) {\rm d}n
\end{aligned} \tag{3.6}$$

Taking the ${\rm d}/{\rm d}s$ derivative outside the integral is allowed provided the integrand is zero at the upper limit, which is the reason for the $n > n_{\rm e}$ requirement for the final relation (3.6). The mass defect is the difference in the mass flow between the EIF and the real flow, integrated across the shear layer, and the displacement thickness is the resulting physical displacement of the potential flow away from the wall.

It must be stressed here that calculation of $m(s)$ and $\delta^*(s)$ requires an analysis of the boundary layer itself, which will be treated in Chapter 4. Here they are assumed to be known properties of the boundary layer.

3.3 Improved Inviscid Flow Models

Two improved EIF models considered next are sketched in Figure 3.3. They define the EIF such that it satisfies the normal mass flux matching requirement (3.5), and thus they capture the displacement effect. The result is much better flow-field prediction accuracy, especially for flows with thick boundary layers.

3.3.1 Displacement Body model

This model employs the concept of a fictitious *displacement body*, which is offset from the actual body by some distance $\Delta n(s)$. The EIF is defined to be tangent to the displacement body, and hence can be constructed by a vortex sheet placed on this displacement body, as shown in Figure 3.3, rather than on the

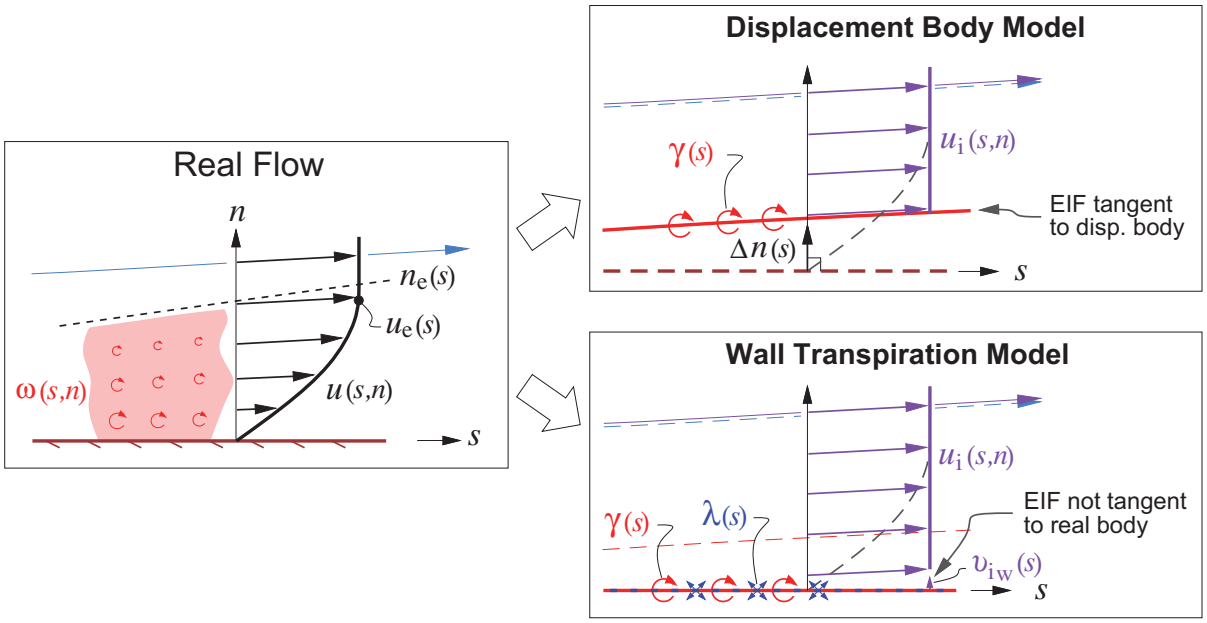

Figure 3.3: Real viscous flow approximated by two improved EIF models which capture the real flow's displacement effect. This mostly eliminates the modeling discrepancies shown in Figure 3.1.

wall as in Figure 3.1. The objective here is to determine what $\Delta n(s)$ has to be so that the vertical mass flux matching condition (3.5) is satisfied.

The EIF's $\rho_i v_i$ in this situation is computed using the continuity equation, as for the real-flow case. Note that $\rho_i v_i$ is not zero at the displacement body, since displacement body's normal vector is tilted away from the n axis by the slope $\mathrm{d}\,\Delta n/\mathrm{d}s$. The model is also assumed a priori to give the correct EIF which matches the real flow, so that we can set $u_i = u_e$.

$$\begin{aligned}
\rho_i v_{i(s,n)} &= \rho_e u_e \frac{\mathrm{d}\,\Delta n}{\mathrm{d}s} + \int_{\Delta n}^{n} \frac{\partial\,\rho_i v_i}{\partial n}\,\mathrm{d}n = \rho_e u_e \frac{\mathrm{d}\,\Delta n}{\mathrm{d}s} - \int_{\Delta n}^{n} \frac{\partial\,\rho_i u_i}{\partial s}\,\mathrm{d}n \\
&= \rho_e u_e \frac{\mathrm{d}\,\Delta n}{\mathrm{d}s} - (n - \Delta n)\,\frac{\mathrm{d}\,\rho_e u_e}{\mathrm{d}s} \\
\text{or} \qquad \rho_i v_{i(s,n)} &= \frac{\mathrm{d}(\rho_e u_e\,\Delta n)}{\mathrm{d}s} - n\frac{\mathrm{d}\,\rho_e u_e}{\mathrm{d}s} \qquad \text{(displacement-body model)} \qquad (3.7)
\end{aligned}$$

Requiring this $\rho_i v_i$ to be equal to the real flow's ρv as defined by (3.6), gives

$$\boxed{\Delta n(s) \;=\; \delta^*(s)} \qquad (3.8)$$

so that the necessary offset for the displacement body is just the displacement thickness (hence the name).

3.3.2 Wall Transpiration model

This model places the vortex sheet on the actual body as in the Simple Inviscid Model shown in Figure 3.1. But now a source sheet of some strength $\lambda(s)$ is also added, as shown in Figure 3.3. This generates a fictitious *wall transpiration* or mass flux distribution $(\rho_i v_i)_{w(s)}$. The resulting EIF is thus intentionally <u>not</u> made tangent to the real body, which enables simulating the displacement effect.

The resulting $\rho_\mathrm{i} v_\mathrm{i}$ at some distance n above the wall again follows from continuity.

$$\rho_\mathrm{i} v_\mathrm{i}(s,n) \;=\; (\rho_\mathrm{i} v_\mathrm{i})_\mathrm{w} \;+\; \int_0^n \frac{\partial\, \rho_\mathrm{i} v_\mathrm{i}}{\partial n}\,\mathrm{d}n \;=\; (\rho_\mathrm{i} v_\mathrm{i})_\mathrm{w} \;-\; \int_0^n \frac{\partial\, \rho_\mathrm{i} u_\mathrm{i}}{\partial s}\,\mathrm{d}n$$

$$\text{or} \qquad \rho_\mathrm{i} v_\mathrm{i}(s,n) \;=\; (\rho_\mathrm{i} v_\mathrm{i})_\mathrm{w} \;-\; n\frac{\mathrm{d}\,\rho_\mathrm{e} u_\mathrm{e}}{\mathrm{d}s} \qquad \text{(wall-transpiration model)} \tag{3.9}$$

Again requiring this to be equal to ρv of the actual flow (3.6) gives the the required wall mass flux.

$$\boxed{(\rho_\mathrm{i} v_\mathrm{i})_\mathrm{w} \;=\; \frac{\mathrm{d}m}{\mathrm{d}s}} \tag{3.10}$$

In a low-speed flow the normal velocity can be imposed using a source sheet, as indicated in Figure 3.3. In this case the airfoil's interior velocity can still be set to zero when the vortex sheet strength $\gamma(s)$ is calculated by the panel method. The required source sheet strength is then equal to the transpiration velocity, and related to the mass defect as follows.

$$\lambda(s) \;=\; (v_\mathrm{i})_\mathrm{w} \;=\; \frac{1}{\rho}\frac{\mathrm{d}m}{\mathrm{d}s} \tag{3.11}$$

3.3.3 Wake modeling

The boundary layers on a body merge together at the trailing edge or rear point and trail downstream as a wake. A wake has two edges — upper and lower — where the potential-flow quantities will now be denoted by $()_u$ and $()_l$, respectively. The particular vertical location of the s-axis, or equivalently the $n=0$ point, is not significant, and it can lie anywhere in or near the wake.

If the wake is thin compared to the streamwise radius of curvature, then it has a nearly uniform static pressure across it like a boundary layer. Also, the lower and upper potential flows have the same freestream total pressure, so that the two edge velocities must also be the same, and so both can be denoted by u_e.

$$p_u(s) \;=\; p_l(s) \qquad \rightarrow \qquad u_u(s) \;=\; u_l(s) \;=\; u_\mathrm{e}(s) \tag{3.12}$$

The 2D wake has a nonzero mass defect and in general nonzero normal mass fluxes at n_u and n_l. Repeating the previous mass flux analysis for the wake, the real-flow edge mass flux relation (3.6) becomes a *jump* condition across the wake.

$$\Delta(\rho v) \;\equiv\; \rho v(s,n_u) \;-\; \rho v(s,n_l) \;=\; \frac{\mathrm{d}m}{\mathrm{d}s} \;-\; (n_u - n_l)\frac{\mathrm{d}\,\rho_\mathrm{e} u_\mathrm{e}}{\mathrm{d}s} \qquad \text{(actual wake)} \tag{3.13}$$

$$\text{where} \qquad m(s) \;\equiv\; \int_{n_l}^{n_u} (\rho_\mathrm{e} u_\mathrm{e} - \rho u)\,\mathrm{d}n \;=\; \rho_\mathrm{e} u_\mathrm{e} \delta^*$$

$$\delta^*(s) \;\equiv\; \int_{n_l}^{n_u} \left(1 - \frac{\rho u}{\rho_\mathrm{e} u_\mathrm{e}}\right)\mathrm{d}n$$

Note that the m and δ^* definitions are the same as for the wall case, aside from the different edge limits.

Repeating the mass flux analysis for the Displacement-Body and Wall-Transpiration models, and requiring that the resulting $\Delta(\rho v)$ matches the real-flow result (3.13), gives the required displacement body thickness, and the required source sheet strength.

$$\Delta n(s) \;=\; \delta^*(s) \qquad \text{(2D-wake Displacement Body model)} \tag{3.14}$$

$$\lambda(s) \;=\; \frac{1}{\rho}\frac{\mathrm{d}m}{\mathrm{d}s} \qquad \text{(2D-wake Wall Transpiration model)} \tag{3.15}$$

These are the same as (3.8) and (3.11) for the boundary layer, except that Δn now only gives the thickness of the wake displacement body. The camber shape of the wake displacement body must be implicitly determined from the zero pressure jump or velocity jump requirement, such that the wake displacement body carries no lift.

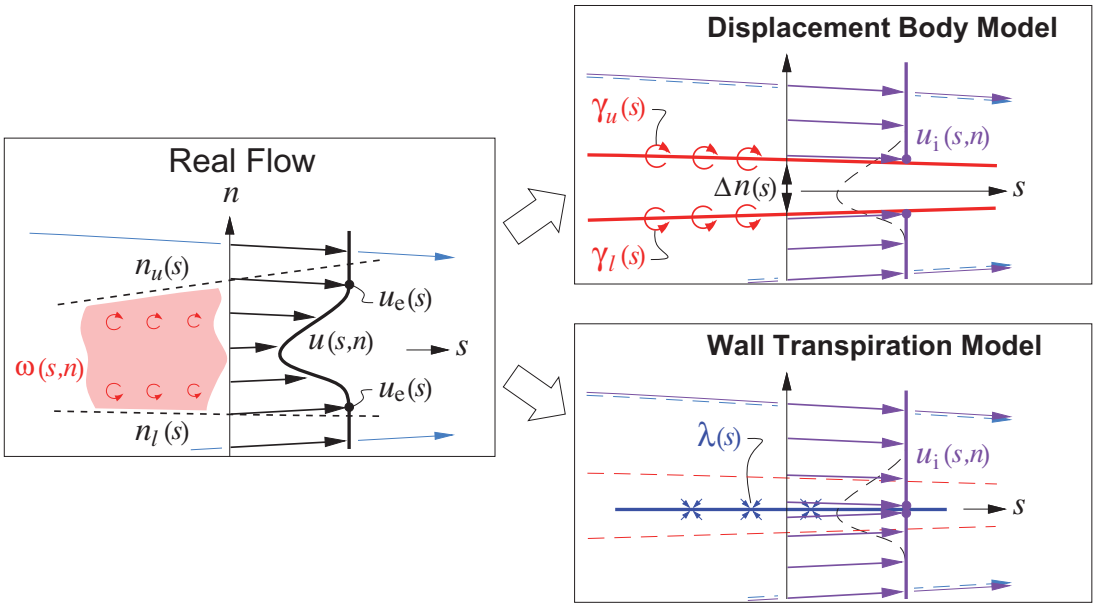

Figure 3.4: Real viscous flow of wake approximated by two EIF models which capture the wake's displacement effect.

The EIF models resulting from the above matching relations are shown in Figure 3.4. Note that the Displacement Body model requires the use of two vortex sheets, which must have equal and opposite strengths as required by zero velocity jump requirement across the whole wake.

$$\gamma_u(s) + \gamma_l(s) = 0 \tag{3.16}$$

3.3.4 Improved flow model advantages

Both the Displacement Body and the Wall Transpiration models quantitatively give very nearly the same results when incorporated into potential flow calculation methods, and both are great improvements over the Simple Inviscid model when separation is present. An example comparison is shown in Figure 3.5 for an airfoil from zero lift to beyond stall. At small lift coefficients where the displacement effects are weak, the three models give comparable results, as shown in Figure 3.6 for $\alpha = 0°$. At a large lift coefficient with trailing edge separation, the differences are quite significant, as shown in Figure 3.7 for $\alpha = 16°$.

3.4 Viscous Decambering Stall Mechanism

The Displacement Body model combined with Glauert's thin airfoil theory [8], sections D.2 and D.3, provides an intuitive explanation for the loss of lift at stall. This theory gives the general lift result

$$c_\ell(\alpha) = 2\pi\alpha + c_{\ell_0} \tag{3.17}$$

for any thin airfoil. The lift intercept c_{ℓ_0} depends only on the airfoil's camberline shape, and is most sensitive to the deflection angle of the camberline over the rear portion of the airfoil. These results also apply to more general airfoils, except with finite thickness the slope $\mathrm{d}c_\ell/\mathrm{d}\alpha = 2\pi$ will be somewhat larger.

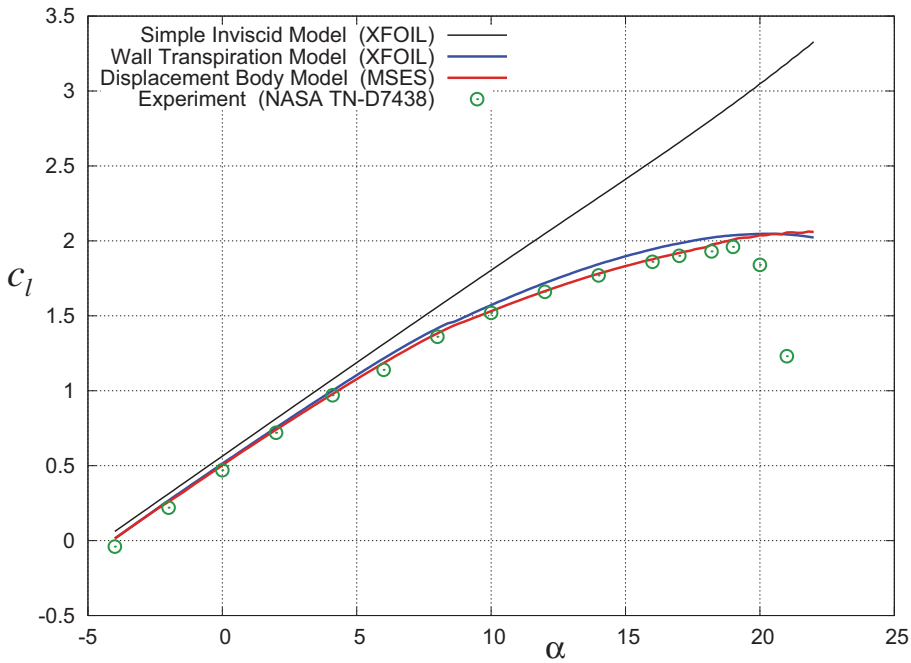

Figure 3.5: Comparison of $c_{\ell(\alpha)}$ lift curves for GAW(1) airfoil at $Re_c = 6\,\text{million}, M_\infty = 0.15$, calculated by XFOIL [5] and MSES [6]. Experimental data is from McGhee et al [7].

In light of this result we examine the camberlines of the displacement bodies of the GAW(1) airfoil at the $\alpha = 0°$ and $\alpha = 16°$ operating points, shown in Figure 3.8. The large separation region over the upper rear of the airfoil at high α increases the displacement body offset Δn there, creating an upward deflection in its camberline which then causes the downward shift in the lift curve for that shape. This upward deflection of the effective camberline is called the *viscous flap*. The two resulting fixed-camberline $c_{\ell(\alpha)}$ functions intersect the actual $c_{\ell(\alpha)}$ curve for the airfoil, which in effect goes through a progression of ever-increasing upward viscous flap deflections which gradually reduce the lift from its strictly inviscid value. Maximum lift and subsequent stall occurs when this viscous flap decambering progression overpowers the inviscid lift coefficient gradient $\mathrm{d}c_\ell/\mathrm{d}\alpha$.

3.5 Considerations in Flow Model Selection

The Simple Inviscid model shown is Figure 3.1 is attractive because it requires only the body geometry and the freestream velocity as inputs. Hence, it's relatively simple to apply via any inviscid-flow theory or calculation method. For this reason it is often used as a first estimate, and whenever the viscous effects can be considered negligible, as for the small angles of attack in the GAW(1) airfoil example above.

Although the Displacement Body or Wall Transpiration models are clearly superior for all cases, they are considerably more complex to implement and to use. The major complication in their implementation is that they require knowledge of the boundary layer's mass defect $m(s)$ or displacement thickness $\delta^*(s)$ distributions. These quantities not known a priori, but must come from an analysis of the boundary layer itself (as treated in Chapter 4). Furthermore, the inviscid flow calculation and the boundary layer calculation are also coupled, in that the output of one is the input to the other. For this reason they must be solved in a coupled manner, as for example in the XFOIL 2D airfoil code [5]. This treatment is considerably more complex than the straightforward use of only an inviscid method, such as a panel method, in the Simple Inviscid model, especially for 3D flows. Section 4.12 gives further discussion of the coupling of inviscid and boundary layer flows.

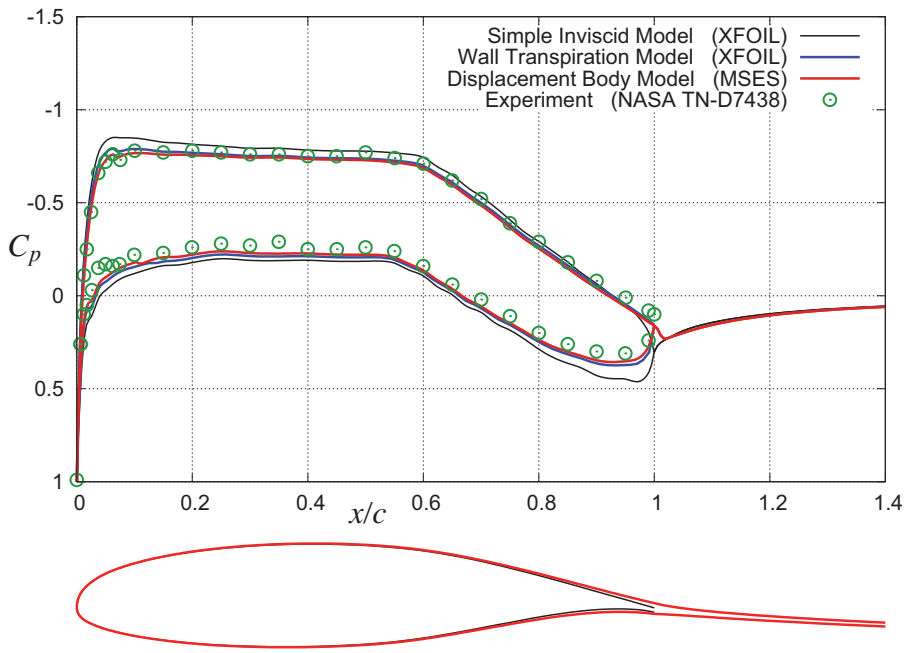

Figure 3.6: Comparison of GAW(1) airfoil surface C_p distributions for the $\alpha = 0°$ point. The airfoil is shown with the displacement body superimposed. The flow is well attached, so that the displacement effects are weak and the displacement body differs only slightly from the actual airfoil.

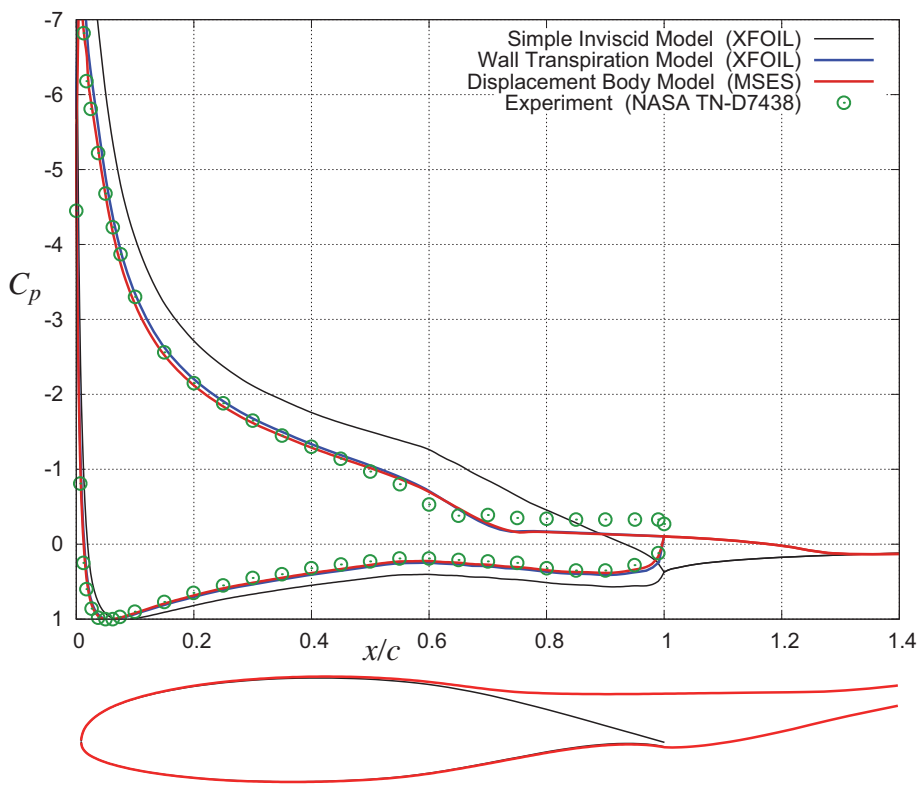

Figure 3.7: Comparison of GAW(1) airfoil surface C_p distributions for the $\alpha = 16°$ point. The displacement body is now very different from the actual body, due to trailing edge boundary layer separation which also extends into the wake.

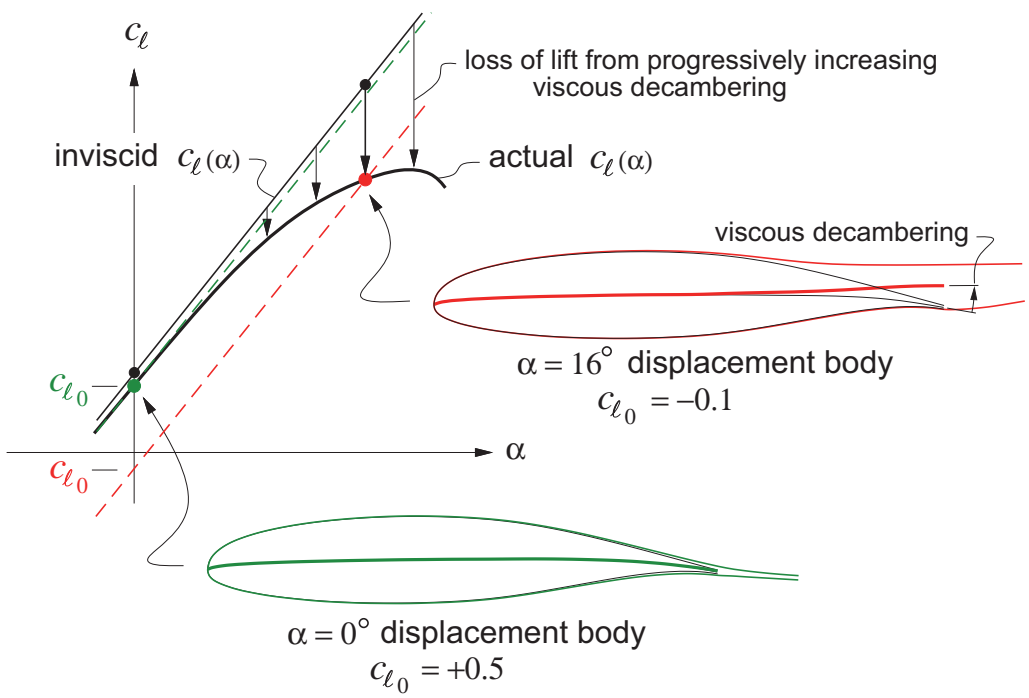

Figure 3.8: Lift curves (dashed) of two GAW(1) displacement bodies with different camberline rear deflections. Viscous decambering correctly models the loss of lift and stall at high angles of attack.